

Ahilesh Vadivel

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EDUCATION

M.S. Robotics | Northeastern University, Boston, MA Expected May 2026
GPA: 4.0/4.0 | Concentration: Electrical and Computer Engineering | *Coursework*: Mobile Robotics, Robot Mechanics & Control, Robot Sensing & Navigation, Robotic Science & Systems

B.E. Electrical and Communication Engineering | Anna University, Chennai, India May 2024
Coursework: Control Systems, Microprocessors & Microcontrollers, Advanced Microcontrollers

TECHNICAL SKILLS

Languages	C++, Python, Embedded C, MATLAB, G-code
Robotics	ROS, ROS 2, RViz, Navigation Stack, Gazebo, MuJoCo, CoppeliaSim
Hardware	PWM actuation, motor drivers (L298N), Arduino (UNO, DUE), Raspberry Pi, PIC microcontrollers, custom PCB design, UR3 arm
Sensing	LiDAR (2D/3D), cameras, IMU, encoders, capacitive & pressure sensors, motion capture
ML & CV	PyTorch, TensorFlow, OpenCV, CNNs, ResNet18, PPO (RL), transfer learning, feature extraction
Tools	Git, Linux/Unix, TensorBoard, PySerial, L ^A T _E X

WORK EXPERIENCE

Graduate Research Assistant | PARSES Lab, Northeastern University | Boston, MA Jan 2025 – Dec 2025

- Developed closed-loop ROS 2 control for Arduino-controlled pneumatic actuators on a UR3 arm; implemented PWM pressure modulation for coordinated multi-finger actuation achieving **90% grasp success rate** across hardware trials.
- Built multi-sensor fusion pipeline synchronizing capacitive, pressure, and motion capture streams via ROS 2 for reliable real-time state estimation; debugged node timing and communication failures through iterative hardware testing.
- Co-developed PID controller for vacuum pressure regulation; validated through 10+ test protocols achieving **<5mm position error (2D), <10mm (3D)**, 680g payload — published in *Soft Robotics Journal*.
- Pioneered Kresling origami soft gripper application to in-hand manipulation; iterated 3- to 4-finger designs with UR3 integration; implemented screwing algorithm via alternating pressure actuation — **85% success rate** across 20+ trials.

Research Student | CSIR – Central Scientific Instruments Organisation | Chennai, India Jul 2022 – Aug 2022

- Designed and validated a multi-sensor monitoring system with custom PCB integrating voltage/current measurement circuitry for characterization of a Peltier-based portable vaccine storage prototype.

PROJECTS

Terrain-Aware Visual Perception for Quadruped Locomotion | Northeastern University Feb – Apr 2026

Python • PyTorch • ResNet18 • MuJoCo • PPO • ROS 2

- Trained ResNet18 on 6,000 synthetic MuJoCo images (6 terrain classes) achieving **100% accuracy at 2.19ms**; extended RL observation space (105→111-D) with terrain vector; improved sim-to-real texture-class accuracy from 65% to 100% via domain randomization.

Reinforcement Learning for Quadruped Locomotion | Northeastern University Feb – Apr 2026

Python • PyTorch • Gymnasium (Ant-v5) • MuJoCo • PPO

- Implemented PPO policy processing 105-D proprioceptive state to output 8 continuous torques; achieved reward of 1,873 on flat terrain via systematic reward shaping for robust legged locomotion in MuJoCo simulation.

SLAM-Based Multi-Robot Autonomous Navigation | Northeastern University Oct – Dec 2024

ROS • Gmapping • AMCL • move_base • Gazebo • RViz • Python

- Configured full ROS navigation stack (Gmapping SLAM, AMCL, move_base) for two simultaneous TurtleBot3 robots in Gazebo; developed ROS node to fuse both occupancy maps into a unified real-time master map.

Camera Calibration & Augmented Reality System | Northeastern University Feb – Mar 2026

Python • OpenCV • Zhang's Method • solvePnP

- Built complete Zhang's calibration pipeline (54 corners, **1.28px RMS reprojection error**); implemented real-time 6-DOF pose estimation with solvePnP for live AR virtual object overlay on video stream.

Vision-Based Pick-and-Place CNC Robot | Anna University Nov 2023 – Feb 2024

Python • CNN • OpenCV • Raspberry Pi • PySerial • G-code

- Trained CNN achieving **99% picking accuracy** across 3 object classes; wrote Python software to translate CNN outputs into G-code commands over serial (PySerial) for fully autonomous pick-and-place operation.

RELEVANT COURSEWORK

Robot Mechanics & Control • Mobile Robotics • Robot Sensing & Navigation • Robotic Science & Systems • Pattern Recognition & Computer Vision • Neural Network Verification for Safety-Critical Systems